Maxwell B. Morrison

Electrical Engineering | Embedded Systems | Signal Processing
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Education

University of Delaware Newark, DE
B.E.E. Electrical Engineering; B.S. Applied Mathematics Expected May 2027
Minor in Computer Science | GPA: 3.55/4.00

Coursework: Digital Systems, Signals and Systems, Digital Signal Processing, Controls, Electromagnetics, Linear Algebra, Probability and Statistics.

Engineering Experience

University of Delaware 2026–Present
Undergraduate Lab Manager

- Manage an undergraduate engineering laboratory supporting electronics, embedded systems, instrumentation, and student hardware design projects.
- Maintain equipment and inventory; assist students with instrumentation, microcontrollers, hardware troubleshooting, 3D printing, and rapid prototyping.

Henry Adams, LLC May 2025–Aug. 2025
Electrical Engineering Intern / MEP Design

- Developed detailed electrical drawings for contractors using Revit and AutoCAD.
- Performed lighting simulations using AGi32 and contributed to arc-flash studies using SKM PowerTools.

Teaching Experience

University of Delaware — ELEG205 Analog Circuits Jan. 2025–Present
Lab Assistant

- Support laboratory instruction, grading, and individual student assistance in undergraduate electrical engineering coursework.
- Help students apply analog circuit concepts, make measurements with engineering instrumentation, and troubleshoot laboratory hardware.

University of Delaware — CPEG202 Digital Logic Jan. 2025–Present
Teaching Assistant / Supplemental Instruction Leader

- Support two course sections through laboratory supervision, grading of student work and exams, and individual assistance.
- Facilitate peer-led problem-solving sessions and help students apply digital logic concepts and debug hardware.

Awards & Leadership

Awards: Research Day People's Choice Award — DroneEye; IEEE Delaware Bay Section Student Activities Award; Dean's List.

Leadership & Activities: Treasurer, IEEE University of Delaware; Eta Kappa Nu (HKN); Eagle Scout, Troop 97.

Technical Skills

Programming: Python, C, C++, MATLAB, Julia, LaTeX; Git
Embedded / Hardware: ESP32, STM32, SPI, I^2C, ADCs, sensors, SD-card storage, embedded systems
ML / Data: PyTorch, scikit-learn, NumPy, pandas, DINOv2, classification, probability calibration, kernel methods, statistical analysis
Engineering Tools: Revit, AutoCAD, AGi32, SKM PowerTools, Fusion 360, Linux

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Research Experience

Ocean Acoustics and Engineering Laboratory Jan. 2025–Aug. 2025
Undergraduate Researcher *University of Delaware*

- Designed and programmed an ESP32-based underwater acoustic data logger for autonomous hydrophone measurements, in collaboration with Professor Mohsen Badiey's research team.
- Implemented 44.1 kHz ADC sampling and binary audio storage on an SD card using C/C++, connecting acoustic measurements with a microcontroller-based acquisition pipeline.
- Integrated DS18B20 temperature sensing through OneWire, collecting environmental measurements at 10-second intervals alongside acoustic data.
- Added RTC timestamping, periodic file creation, and separate audio and temperature output files to organize unattended measurements for later analysis.
- Troubleshot SD-card mounting, SPI pin assignments, empty output files, and temperature-data formatting during hardware and firmware integration.
- Integrated a lithium-ion battery and 3.3 V regulation with acoustic acquisition, environmental sensing, and removable storage for autonomous operation.

Computational Neural Information Engineering Laboratory Summer 2026
Undergraduate Researcher *University of Delaware*

- Developed semi-supervised machine-learning methods that select unlabeled samples to augment labeled training data, with research advised by Professor Austin Brockmeier.
- Extracted 384-dimensional image embeddings using DINOv2 ViT-S/14 with 224-by-224 image inputs, using pretrained representations as classifier features.
- Constructed shifted-cosine kernel similarity matrices and normalized each matrix by its trace to form the density matrix used in spectral entropy calculations.
- Used von Neumann entropy and a scaled log-probability penalty to balance sample diversity with confidence in class assignments.
- Built Python pipelines for feature extraction, one-versus-rest logistic regression, Platt probability calibration, sample selection, and model retraining.
- Compared entropy-guided augmentation against labeled-only and random-selection baselines across repeated experiments; evaluated accuracy, precision, recall, F1, and ROC AUC.
- Co-authored research on fixed-cardinality kernel similarity matrices for entropy-guided semi-supervised data augmentation and presented findings in research presentations.
- Evaluated selection penalties, trust weights, and logistic-regression regularization across 10 random seeds to assess performance and variability relative to baseline methods.
- Conducted experiments with 1,224 initial training samples, 2,448 samples after augmentation, and a separate 10,000-sample test set.
- Presented research on maximizing von Neumann entropy for semi-supervised data augmentation at University of Delaware Research Day.

Projects

DroneEye — Drone Detection and Localization Fall 2025–Spring 2026
Undergraduate Project Team Member *University of Delaware*

- Collaborate with an undergraduate team to develop a LiDAR-assisted computer vision system for drone detection and Cartesian-coordinate estimation.
- Incorporate a graphical user interface into the system to support interaction with detection and localization outputs.
- Present DroneEye at University of Delaware Research Day; project receives the People's Choice Award.

PitchMAX — Pitch-Calling System
Hardware and Firmware Project

- Developed a lower-cost pitch-calling system inspired by PitchCom, using custom hardware and firmware to communicate called pitches between players.

PyroPlan — Fire-Spread Simulation Platform
Web Application / Simulation Project

- Created a web platform for modeling fire spread in level 1 materials, with simulations performed server-side.
- Made the simulation accessible through a web interface without requiring users to install local simulation software.

Diabeasy — Insulin Tracking Application
Python / Hackathon Project

- Built an encrypted Python application using Tkinter to help users track insulin usage and calculate dosage amounts.